

# IS MATTER AROUND US PURE

Class 9 | Science | Chapter 2

## MIXTURES

A mixture contains two or more substances (elements/compounds) mixed in ANY ratio, not chemically combined.

Ex: Air, sea water, soil, tea, alloys like brass.

### Homogeneous Mixture

Uniform composition throughout. Ex: sugar solution, air, alloys.

### Heterogeneous Mixture

Non-uniform composition; components visible. Ex: sand + water, oil + water.

## SOLUTIONS

A solution is a homogeneous mixture of two or more substances.

- **Solute** -- the substance dissolved (smaller amount). **Solvent** -- the substance that dissolves it (larger amount).
- Ex: Sugar (solute) + Water (solvent) = Sugar solution.
- A solution has particles smaller than 1 nm -- so it does NOT scatter light and is stable (does not settle).

### Concentration of a Solution

$\text{Mass \%} = (\text{Mass of solute} / \text{Mass of solution}) \times 100$ . Concentration = amount of solute present in a given amount of solution.

## TYPES OF SOLUTIONS

| Type                   | Example                               |
|------------------------|---------------------------------------|
| Solid in liquid        | Salt in water                         |
| Gas in liquid          | Soda water (CO <sub>2</sub> in water) |
| Liquid in liquid       | Vinegar (acid in water)               |
| Solid in solid (alloy) | Brass (Cu + Zn)                       |
| Gas in gas             | Air                                   |

### Saturated vs Unsaturated

*Saturated: no more solute can dissolve at that temperature. Unsaturated: more solute can still dissolve. Solubility generally increases with temperature.*

# SUSPENSIONS, COLLOIDS & SEPARATION

Comparison + Tyndall Effect

## SUSPENSION

Heterogeneous mixture where particles are large enough (visible to naked eye) to scatter light and settle down on standing.

Ex: Muddy water, chalk in water, paint.

## COLLOID (COLLOIDAL SOLUTION)

Heterogeneous mixture that APPEARS homogeneous; particles are too small to see but large enough to scatter light (Tyndall Effect).

Ex: Milk, fog, smoke, blood, shaving cream, ink.

### DIAGRAM: LIGHT PATH IN EACH TYPE

#### SOLUTION



No scattering -- light passes through

#### COLLOID



Tyndall Effect -- scatters light path visible

#### SUSPENSION



Blocks light, particles settle on standing

## SOLUTION VS COLLOID VS SUSPENSION

| Property       | Solution           | Colloid                       | Suspension          |
|----------------|--------------------|-------------------------------|---------------------|
| Particle size  | < 1 nm             | 1-1000 nm                     | > 1000 nm           |
| Appearance     | Transparent        | Translucent                   | Opaque              |
| Settling       | Does not settle    | Does not settle               | Settles on standing |
| Tyndall effect | No                 | Yes                           | Yes (sometimes)     |
| Filterability  | Cannot be filtered | Cannot be filtered (ordinary) | Can be filtered     |

# SEPARATION TECHNIQUES

Physical Methods to Separate Mixtures

## COMMON SEPARATION METHODS

| Technique               | Used to separate                                                                                         |
|-------------------------|----------------------------------------------------------------------------------------------------------|
| Evaporation             | Solid dissolved in liquid (e.g. salt from water)                                                         |
| Filtration              | Insoluble solid from liquid (e.g. sand from water)                                                       |
| Distillation (simple)   | Liquid from a solution with dissolved solid; or two miscible liquids with large boiling point difference |
| Fractional distillation | Two or more miscible liquids with close boiling points (e.g. petroleum, mixture of gases from air)       |
| Sublimation             | A sublimable solid (ammonium chloride, camphor, naphthalene) from a non-sublimable one                   |
| Chromatography          | Separating dyes/pigments that are dissolved in the same solvent                                          |
| Separating funnel       | Two immiscible liquids (e.g. oil and water)                                                              |
| Centrifugation          | Very fine suspended particles (e.g. cream from milk, blood components)                                   |
| Magnetic separation     | A magnetic component from a mixture (e.g. iron filings from sulphur)                                     |

## CRYSTALLISATION

**A process to purify a solid; obtains pure crystals of a substance from its impure solution -- better than evaporation as it avoids decomposition of the substance.**

*Ex: Purification of impure sugar/copper sulphate.*

# PURE SUBSTANCES & SUMMARY

Elements, Compounds + Quick Revision

## PURE SUBSTANCE

A single kind of particle throughout; cannot be separated by physical methods. Made of only ONE type of element or compound.

## ELEMENT

*Simplest form of matter that cannot be split into simpler substances by chemical reactions.*

| Type       | Examples                           |
|------------|------------------------------------|
| Metals     | Iron, Copper, Gold, Sodium         |
| Non-metals | Oxygen, Carbon, Sulphur, Nitrogen  |
| Metalloids | Boron, Silicon, Germanium, Arsenic |

## COMPOUND

Formed when two or more elements combine chemically in a FIXED ratio; properties differ completely from constituent elements.

*Ex: Water (H<sub>2</sub>O), Carbon dioxide (CO<sub>2</sub>), Common salt (NaCl).*

## MIXTURE VS COMPOUND

| Property                 | Mixture             | Compound                  |
|--------------------------|---------------------|---------------------------|
| Composition              | Any ratio           | Fixed ratio               |
| Properties               | Same as components  | Different from components |
| Separation               | By physical methods | By chemical methods only  |
| Energy change on forming | No                  | Usually yes (heat/light)  |

## ONE-LINE RECAP

Matter: Pure (element/compound) or Mixture (homogeneous/heterogeneous).

Solution < Colloid < Suspension -- in particle size and stability.

Choose separation technique based on physical property difference.

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